

DO NOT REMOVE FROM EXAM VENUE

University of Plymouth

MODULE CODE: ELEC347

**TITLE OF PAPER: INFORMATION & COMMUNICATION
SIGNAL PROCESSING**

TIME ALLOWED

Three hours

DATE

Friday 20th January 2023

TIME

09:00 AM – 12:00 PM

FACULTY

Science and Engineering

SCHOOL

Engineering, Computing and Mathematics

ACADEMIC YEAR

2022/2023

STAGE

Four

INSTRUCTIONS TO CANDIDATES:

The paper consists of **TWO** Sections.

Section A: Candidates should Attempt **ALL 10 QUESTIONS** in Section A.

Section B: Candidates should attempt any **TWO** questions from the four available in this section. All questions in this Section carry equal marks. Marks for parts of questions are shown where appropriate.

Candidates are not permitted to look at the examination paper until instructed to do so.

SECTION A

Answer **ALL** questions.

1. State and justify the sampling frequency that is used in practice in each of the following applications:
 - (a) Voice telephony.
 - (b) Studio audio signal analysis.

[4 marks]

2. (a) State the bandpass sampling theorem and the necessary condition to sample at the minimum rate possible without aliasing.

[4 marks]

(b) A digital receiver has an intermediate frequency (IF) stage with a bandwidth of 16 kHz and an IF frequency of 744 kHz. Assume the input signal to the stage is centred at the IF frequency, determine the minimum theoretical sampling rate to avoid aliasing. Justify your answer.

[4 marks]

3. The analogue signal input to a real-time DSP system is sampled at 3.072 MHz with an oversampling ratio of 32. Determine the maximum signal frequency of interest in the analogue signal.

[4 marks]

4. The analogue input to a real-time DSP system has a full-scale range of $\pm 2.5V$ and the peak noise present at the input is $60\mu V$. Estimate the minimum number of bits required to digitise the signal so that the peak quantisation noise is not greater than the peak noise at the input. Give details of how you arrived at your answer.

[5 marks]

5. The DFT of a real-valued data sequence is computed using an 8-point, radix-2 FFT processor. If the first 5 DFT coefficients are:

$$X(0) = 27, X(1) = -8 + 6j, X(2) = 8 + 2j, X(3) = 0 - 10j, X(4) = 12.$$

- (a) Write down the values for $X(5)$, $X(6)$ and $X(7)$. Justify your answer.

[3 marks]

- (b) Determine the dc value of the sequence.

[1 mark]

6. A requirement exists for a linear phase FIR low pass filter in an audio system. Assume a sampling rate of 48 kHz and that the filter can introduce a maximum latency of 13mS, determine the maximum number of coefficients the filter can have.

[4 marks]

7. A discrete-time filter has a pair of complex conjugate zeros located at $\pm j$ and a pair of complex conjugate poles at $\pm 0.9j$ in the z-plane.

- (a) Sketch the pole-zero diagram of the filter and deduce its transfer function.

[5 marks]

- (b) Sketch the magnitude frequency response between 0 and the sampling frequency. State what type of filter is represented (e.g. low pass, high pass, band pass or band stop).

[3 marks]

8. (a) Explain how overflow errors arise in fixed-point digital filters.

[2 marks]

Question 8 Continued ...

Question 8 Continued

- (b) State **TWO** main adverse effects of overflow errors on the performance of digital filters in practice.

[2 marks]

9. A 16-point, radix-2 FFT processor is to be used to analyse a real input data sequence. Determine the bit reverse indices for each of the following input data samples and explain how this is then used in the FFT

(a) $x(1)$

(b) $x(3)$

[4 marks]

10. What is the main disadvantage of oversampling at the analogue I/O interface in practice? How can this be addressed in practice?

[5 marks]

(Over/...)

SECTION B

Answer any **TWO** questions

11. An analogue signal has the spectrum depicted in **Figure Q11a**. Assume the highest frequency components in the signal, $f_{\max} = 20$ kHz, sketch and label the spectrum of the signal after sampling between the range $\pm F_s$ for each of the following cases:

(a) The sampling frequency, F_s , is 30 kHz

[3 marks]

(b) The sampling frequency, F_s , is 40 kHz

[3 marks]

(c) The sampling frequency, F_s , is 80 kHz.

[3 marks]

(d) Compare the results in (a), (b) and (c) above and comment on their practical implications.

[5 marks]

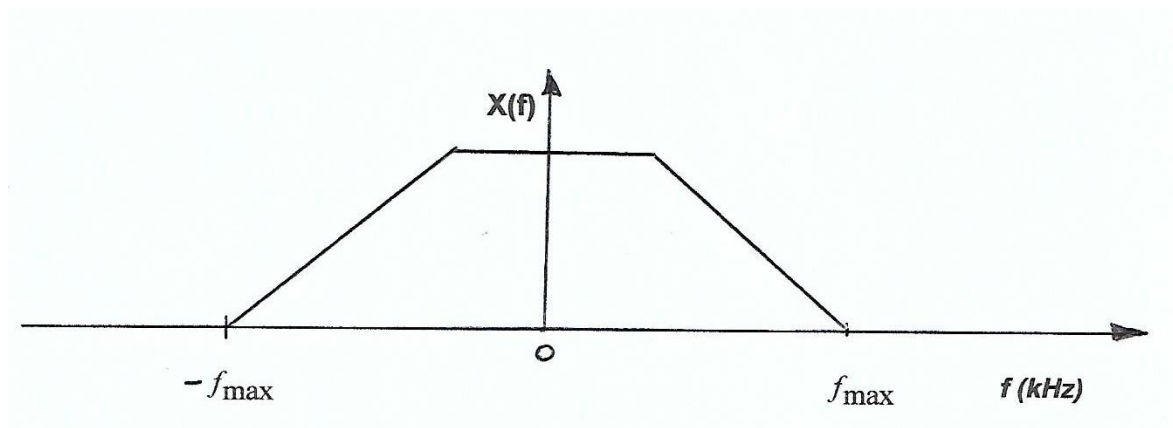


Figure Q11a

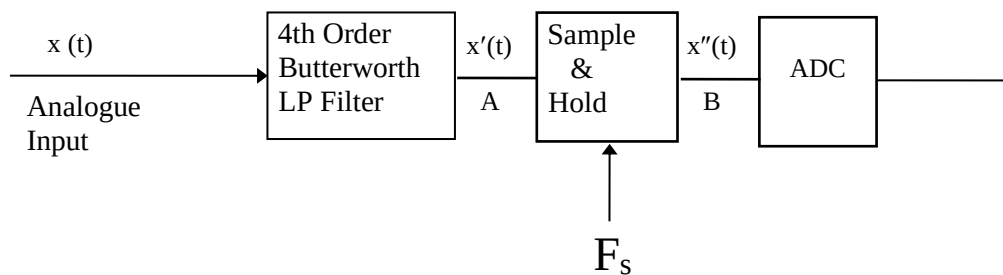
Question 11 Continued...

Question 11 Continued

- (e) The front-end of a real-time DSP system is depicted in the **Figure Q11b**. Assume that the input signal is wide band and that $f_c = 24$ kHz, determine with the aid of appropriate sketches the minimum sampling frequency, F_s (min), to give a signal-to-aliasing error ratio of 60dB at 24 kHz. State any assumptions made.

[11 marks]

Figure Q11b



The normalised amplitude frequency response of an Nth order Butterworth low pass filter with a cutoff frequency, f_c , is given by:

$$H(f) = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^{2N}}}$$

12. (a) **Figure Q12 (page 7)**, shows the flowgraph for an 8-point, radix-2, decimation-in-time FFT processor.

- (i) Use the flowgraph to determine the magnitude-frequency response for a digital filter characterised by the following transfer function:

$$H(z) = h(0) + h(1)z^{-1} + h(2)z^{-2} + h(3)z^{-3}$$

$$\text{where } \{h(k)\} = \{2, 4, 4, 2\}$$

[14 marks]

Question 12(a) Continued...

Question 12(a) Continued

- (ii) Plot and label the magnitude response of the filter. Assume a sampling frequency of 1 kHz. What type of filter is this (low pass, high pass, band pass or band stop filter)? Justify your answer.

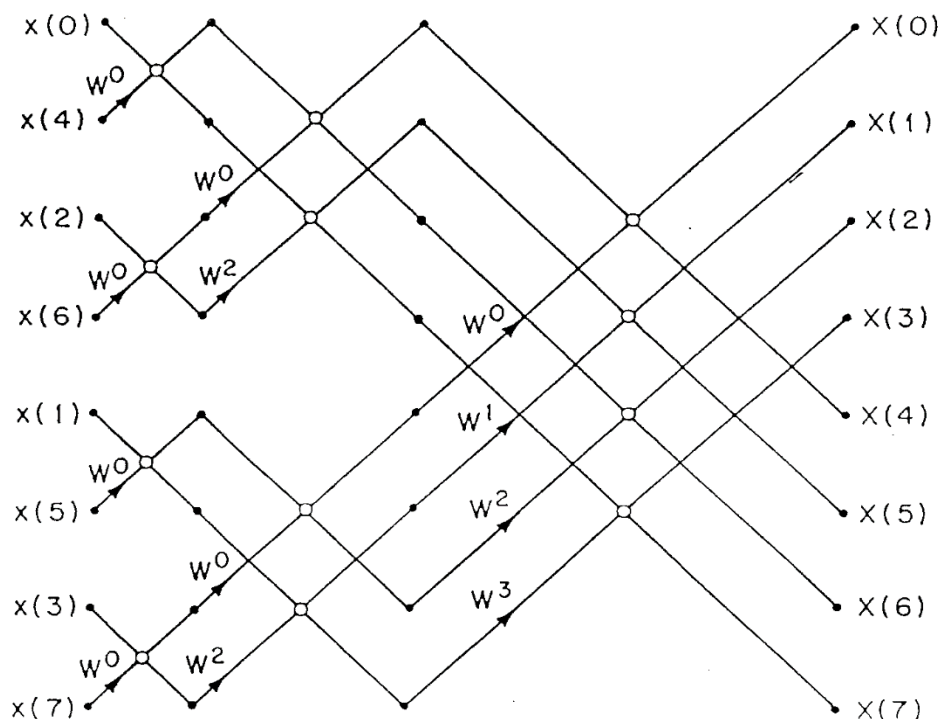
[3 marks]

- (b) An analogue signal to be spectrum analysed using radix-2 FFT is sampled at the rate of 1kHz. Assuming that the signal contains two signal components, spaced 0.5 Hz apart, one of which is 100 times larger than the other.

Determine a suitable window function for the spectrum estimation and the minimum number of actual data points, N , that must be collected.

[8 marks]

NB: You may use the information in **Table Q12 (page 8)**.



$$\text{NB: } W^0 = 1, W^1 = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j, W^2 = -j, W^3 = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j$$

Figure Q12

Question 12 Continued...

Question 12 Continued

Table Q12 Characteristics of window functions in practical use

Window functions	Sidelobe level (dB)	Mainlobe width (Hz)
Rectangular	-13	2/NT
Bartlett	-25	4/NT
Hanning	-31	4/NT
Hamming	-41	4/NT
Blackman	-57	6/NT
Kaiser	Variable	Variable

13. (a) A design requirement exists for a digital bandpass filter with Butterworth characteristics and meeting the following specifications:

Passband	500 – 750 Hz
Sampling frequency	5000 Hz
Filter order	2

Starting with a suitable prototype analogue lowpass filter, determine the transfer function of the digital filter using the bilinear z-transform method.

Justify your choice of prototype filter.

[11 marks]

- (b) Determine and sketch the poles and zeros of the prototype analogue lowpass filter, the pre-warped analogue bandpass filter, and the digital bandpass filter in (a), and hence explain the origin of the poles and zeros of the digital filter.

[7 marks]

- (c) Starting with the transfer function of the digital filter in (a), derive the difference equation of the filter.

[4 marks]

Question 13 Continued...

Question 13 Continued

- (d) Sketch and label the realisation diagram of the digital filter from the difference equation. Assume a direct form realisation structure.

[3 marks]

NB: You may use the filter design information in Table Q13

N.B: (1) The pre-warped band edge frequencies for a bandpass filter are given by:

$$\omega'_{p1} = \tan\left(\frac{\omega_{p1}T}{2}\right); \quad \omega'_{p2} = \tan\left(\frac{\omega_{p2}T}{2}\right)$$

and the lowpass-to-bandpass transformation is given by: $s = \frac{s^2 + \omega_0^2}{Ws}$

where ω_{p1} , ω_{p2} and ω_0^2 are the lower and upper band edge frequencies, and centre frequency respectively. $\omega_0^2 = \omega'_{p1}\omega'_{p2}$, and $W = \omega'_{p2} - \omega'_{p1}$.

(2) The bilinear z-transformation is given by:

$$s \rightarrow \frac{z-1}{z+1}$$

(3) The roots of the second order quadratic equation, ax^2+bx+c , are:

$$R1 = [-b + \text{SQRT}(4ac - b^2)]/(2a)$$

$$R2 = [-b - \text{SQRT}(4ac - b^2)]/(2a)$$

Table.Q.13 Filter design information

14. (a) The block diagram of a single stage sample rate converter is depicted in **Figure Q14a (page 11)** where F_s and F_s/M are the input and output sampling rates, respectively, and M is the decimation factor.

- (i) Explain briefly how the sample rate converter works. Your answer should point out the roles of the digital filter and the sampling rate compressor.

[6 marks]

- (ii) What should the input to the filter be bandlimited to and why?

[2 marks]

Question 14 Continued ...

Question 14 Continued ...

(b) **Figure Q14b (page 11)** shows the signals at the input and output of the sample rate converter, $x(n)$ and $y(m)$, for $M=3$ and at the output of the filter, $w(n)$. From analysis of the signals

- (i) Write down the output samples of the sample rate converter, $y(m)$, in terms of the filter output, $w(n)$, for $m=0, 1, 2, 3$, and 4.

[4 marks]

- (ii) Deduce the general expression for $y(m)$ in terms of $w(n)$.

[3 marks]

(c) A requirement exists for a 2-stage decimator which is to be used to reduce the sampling rate from 3072 kHz to 48 kHz.

- (i) Draw the block diagram of the decimator and indicate the appropriate sampling rate at the output of each decimation stage. Assume a decimation factor pair of 32 and 2.

[4 marks]

- (ii) For the decimator in (i), determine the band edge frequencies for the decimating filter at each stage to avoid aliasing and sketch and label the idealised frequency response. Assume that the band of interest extends from 0 to 20 kHz.

[6 marks]

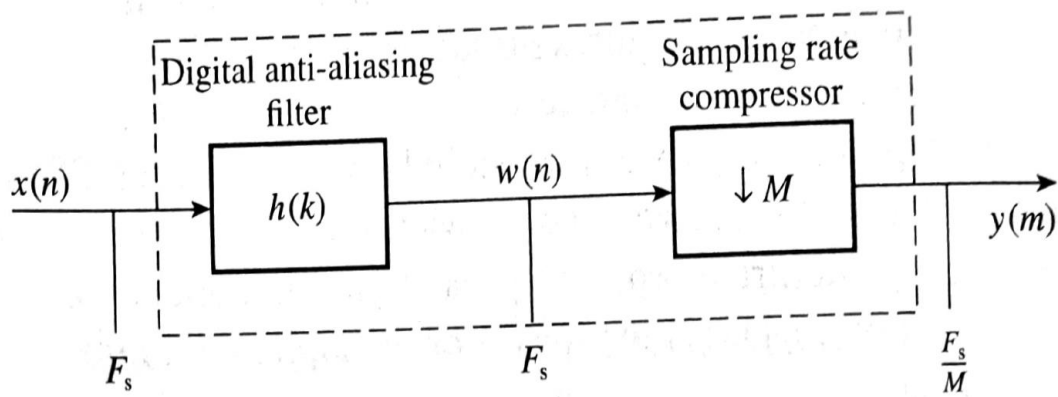
NB: The stop band edge frequency for the decimating filter for the i^{th} stage is given

by: $f_{si} = F_i - \frac{F_s}{2M}$, where F_i is the output sampling rate for the i^{th} stage, F_s is the

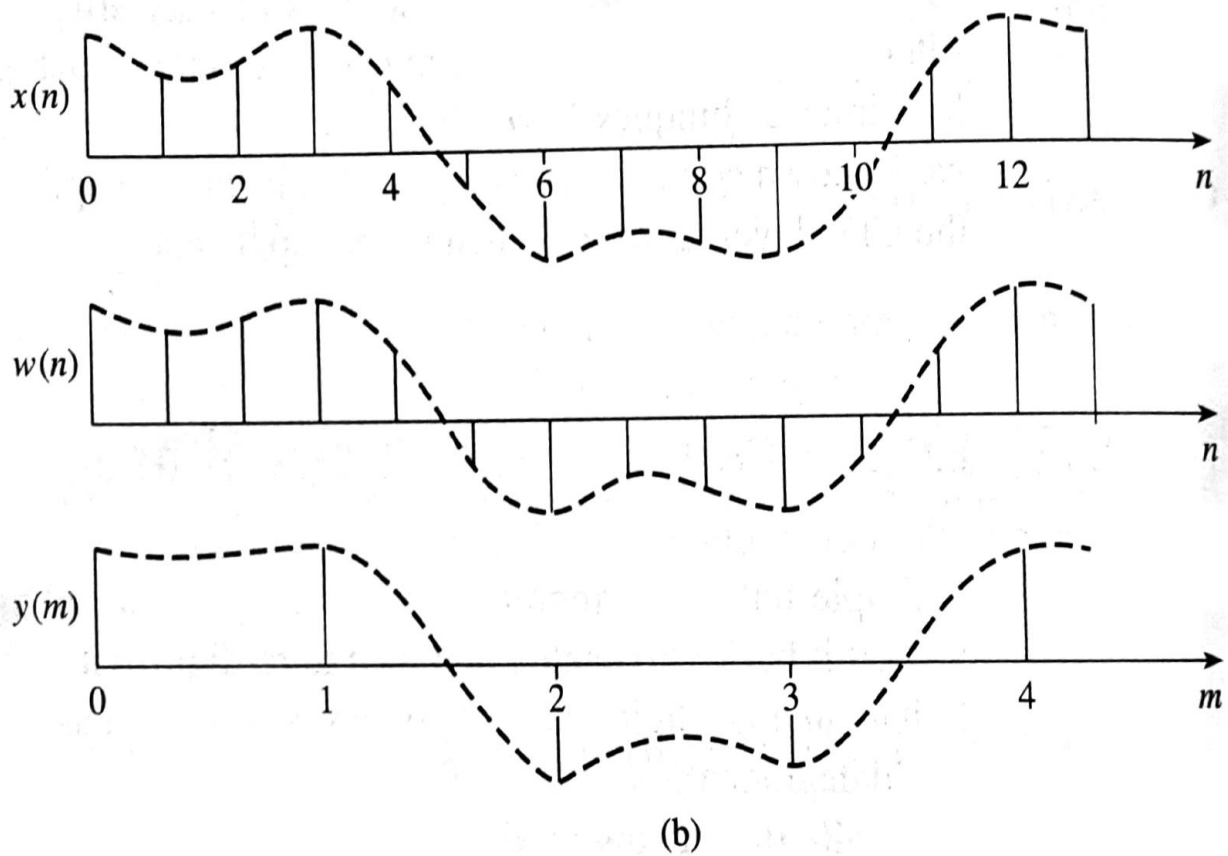
sampling rate for the original data, and M is the overall decimating factor.

Question 14(c) Continued...

Question 14(c) Continued



(a)



(b)

Figure Q14

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